

Eigenvalues and inverse problems

Construct a separating diagonal perturbation in nearly linear time

Difficulty: challenging **Importance:** interesting to the community**Status:** Partially resolved**Topic:** Hermitian eigenproblems; deterministic regularization**Rating rationale:** Challenging reflects derandomizing eigenvalue separation within a nearly linear cost; community impact is deterministic preprocessing for Hermitian eigensolvers.

Partial result — 12 September 2026

Sidney Holden (Center for Computational Biology, Flatiron Institute, Simons Foundation) gives two deterministic constructions in [Theorems A and B](#). Writing $T = \text{diag}(t_i) + R$ and $\eta = \max_i \sum_{j \neq i} |R_{ij}|$, Theorem A covers $\eta \leq \delta/[2(n-1)]$ with gap at least $\delta/(n-1)$ in $O(n \log n)$ operations. Theorem B covers every constant-diagonal tridiagonal input with equal off-diagonal magnitudes, allowing arbitrary complex phases and every coupling strength, with gap at least $\delta/n^3 \geq (\delta/n)^3$ in $O(n)$ operations. Section 5 also covers every order-two input. Each construction returns a real diagonal perturbation of norm at most δ in the original arithmetic model.

These subclasses passed an [independent informal Codex AI-agent audit](#). The all-dimensional Toeplitz starting class suggested in the workshop is included, supporting Partially resolved. Arbitrary normalized Hermitian tridiagonal inputs outside these classes remain open. [Submission and verified affiliation](#); no novelty, external human peer review or formal verification is claimed.

Context and notation

All norms are Euclidean vector norms or their induced matrix norms.

Problem statement

For a Hermitian matrix H with eigenvalues listed with multiplicity, define

$$\text{gap}(H) = \min_{i \neq j} |\lambda_i(H) - \lambda_j(H)|.$$

Do universal constants $a, c_0, C, q > 0$ and a deterministic algorithm exist with the following property? For every $n \geq 2$, Hermitian tridiagonal $T \in \mathbb{C}^{n \times n}$ with $\|T\|_2 \leq 1$, and $\delta \in (0, 1/2)$, the algorithm receives the three diagonals and returns a real diagonal matrix D such that

$$\|D\|_2 \leq \delta, \quad \text{gap}(T + D) \geq c_0(\delta/n)^a,$$

using at most $Cn[1 + \log(n/\delta)]^q$ exact arithmetic operations and comparisons? Square roots of nonnegative real numbers are allowed at unit cost; extracting bits from exact real inputs is not.

This selects the explicitly proposed nearly linear version of the deterministic Minami problem. It preserves tridiagonal structure and requires only the perturbation as output. Its diagonal restriction and cost target make it distinct from the general, cubic-cost regularization question [IE-07](#).

References

Amsel et al., [workshop report](#), Problem 3.2. Sobczyk, [Deterministic complexity analysis of Hermitian eigenproblems](#), §1.1 for the arithmetic model, Theorem 1.1 for the tridiagonal eigensolver, and §5, question 4, for deterministic spectral separation.

Earlier status check — 2026-09-08

Searches for *deterministic Minami perturbation*, *Minami deterministic eigenvalue gap algorithm 2026*, and *Minami derandomization matrix* found no construction meeting this bound. Sobczyk's April 2025 revision gives a deterministic full tridiagonal diagonalization in nearly quadratic time; it does not provide the requested nearly linear diagonal perturbation. The workshop retains Problem 3.2 in its August 2026 update. The normalization and a bounded perturbation are stated explicitly here, making the scale of its gap guarantee unambiguous.

Audit update — 2026-09-10

Rechecked [workshop version 3](#), Problem 3.2. Searches for deterministic Minami-type separation and nearly linear tridiagonal perturbation constructions found no solution of the simultaneous separation and runtime requirements.